

NVH Source Locator —

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NVH Source Locator TDOA (Time Difference of Arrival)

-
-
-
- 2-Sensor
- 3-Sensor
- Pro+ (3-Sen+, 4-Sensor, 4-Sen+, 3D, 3D+)
- Materials
-
-
-
-
- Pro
- Help
-

 {#how-it-works}

NVH Source Locator :

- **Resolution:** The number of pixels in the image (width and height) (e.g., 1920x1080)
- **Frame Rate:** The number of frames per second (FPS) (e.g., 30 FPS)
- **Bit Rate:** The amount of data transferred per second (e.g., 10 Mbps)
- **Compression:** The process of reducing the size of the data (e.g., H.264, H.265)
- **2D:** Two-dimensional (X, Y)
- **3D:** Three-dimensional (X, Y, Z)

Resolution and Frame Rate {#before-you-start}

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NVH

NVH Source Locator

TDOA Calculator PRO

2-Sensor

3-Sensor

3-Sen+

4-Sensor

4-Sen+

3D

3D+

Materials

Pair A-B

Pair A-C

Step 1

Calibration — Speed of Sound

Sensor spacing A-B

-

118

+

cm

External knock time delay

-

471

+

μs

Calculated speed of sound

2505 m/s

Closest: PEEK (2500 m/s)

+ Save as custom material

Step 2

Event Measurement

Event time delay

⊕ trials

-

227

+

μs

Which sensor heard it first?

Sensor A

Calculate Source Location

2-Sensor

{#the-main-tabs}

:

Tab bar (scrolls horizontally on narrow screens)



NVH Source Locator

TDOA Calculator

PRO





2-Sensor

3-Sensor

3-Sen+

4-Sensor

4-Sen+

3D

3D+

Materials

Help

Free

partial

partial

partial

partial

partial

partial

2-Sensor	1	
3-Sensor	2	
3-Sen+	3-Sensor	
4-Sensor	2D	
4-Sen+	2D	4
3D	3	
3D+	3D	6
Materials	+	
Help		

vs Pro: 2-Sensor

Pro

() Pro payoff

()

2-Sensor {#2-sensor-mode}

:

NVH

NVH Source Locator

TDOA Calculator PRO

2-Sensor

3-Sensor

3-Sen+

4-Sensor

4-Sen+

3D

3D+

Materials

Pair A-B

Pair A-C

Step 1

Calibration — Speed of Sound

Sensor spacing A-B

-

118

+

cm

External knock time delay

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471

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μs

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Step 2

Event Measurement

Event time delay

⊕ trials

-

227

+

μs

Which sensor heard it first?

Sensor A

Calculate Source Location

2-Sensor

1:

Materials (

"", "Mild (1020)"

"

2

2:

2-Sensor : A-B A-C (A-B
2)

:

- (d):
()
- (tCal): —

3:

- (tEvent):
- : (A B)

4:

A:

- = 0: A
- = : B
- :
- : (toast)

(A, B)

5 ():

A
 A, B

3-Sensor {#3-sensor-mode}

2D

NVH

NVH Source Locator

TDOA Calculator

PRO

2-Sensor

3-Sensor

3-Sen+

4-Sensor

4-Sen+

3D

3D+

Materials

Same triangle as 3-Sensor, but calibrate & measure **all three pairs** (A-B, A-C, B-C). The solver uses all 3 TDOAs in a least-squares fit – more robust to measurement noise and anisotropic materials. The *RMS residual* tells you how consistent your measurements are.

▲ Show placement guide

Setup

Triangle side lengths

A-B side length

-

100

+

cm

A-C side length

-

90

+

cm

B-C side length

-

80

+

cm

Pair A-B

Calibration & measurement

Calibration time delay – knock outside A-B

-

400

+

μs

2,500 m/s – Closest: PEEK (2,500 m/s)

Event time delay

-

120

+

μs

⊕ trials

Which sensor heard it first?

Sensor A

3-Sensor

■■■■ Pro+ {#pro-modes}

20 °C

Materials

11 12 13 14 15 16 17 18 19 20

[illegible]

- ■■■■■■■■■■■■
- ■■■■■■, Mild (1020)
- ■■■■■■■■■■■■■■■■■■■■ (304)
- ■■■■■■ (■■■■■)
- ■■■■■■
- ■■■■■■
- ■■■■■■■■■■■■
- ■■■■■■

- 
- 
- 
- 
- 
- 

Sensor Type	Count
1-sensor	10
2-sensor	15
3-sensor	8
4-sensor	5
5-sensor	3
6-sensor	2
7-sensor	1
8-sensor	1
9-sensor	1
10-sensor	1
11-sensor	1
12-sensor	1
13-sensor	1
14-sensor	1
15-sensor	1
16-sensor	1
17-sensor	1
18-sensor	1
19-sensor	1
20-sensor	1
21-sensor	1
22-sensor	1
23-sensor	1
24-sensor	1
25-sensor	1
26-sensor	1
27-sensor	1
28-sensor	1
29-sensor	1
30-sensor	1
31-sensor	1
32-sensor	1
33-sensor	1
34-sensor	1
35-sensor	1
36-sensor	1
37-sensor	1
38-sensor	1
39-sensor	1
40-sensor	1
41-sensor	1
42-sensor	1
43-sensor	1
44-sensor	1
45-sensor	1
46-sensor	1
47-sensor	1
48-sensor	1
49-sensor	1
50-sensor	1

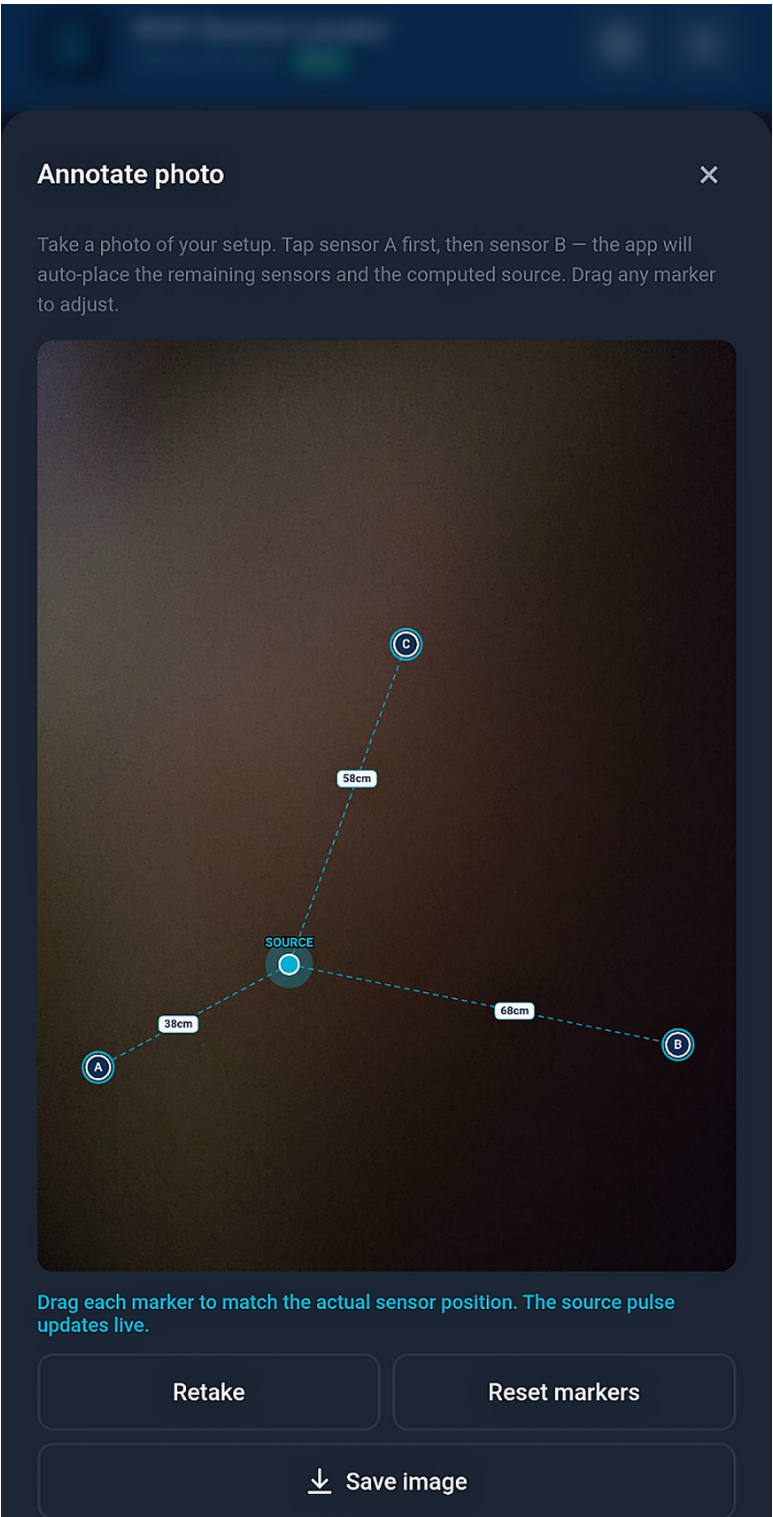
_____;

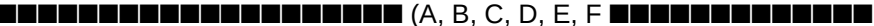
 (_____)

11 12 13 14 15 16 17 18 19 20

□ □ □ □ □ □ □ □

11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26



-  — 
- 
- 
-  (A, B, C, D, E, F  —  6
)

- Android: PDF →
- iOS: → PDF, AirPrint,

→ ,
,

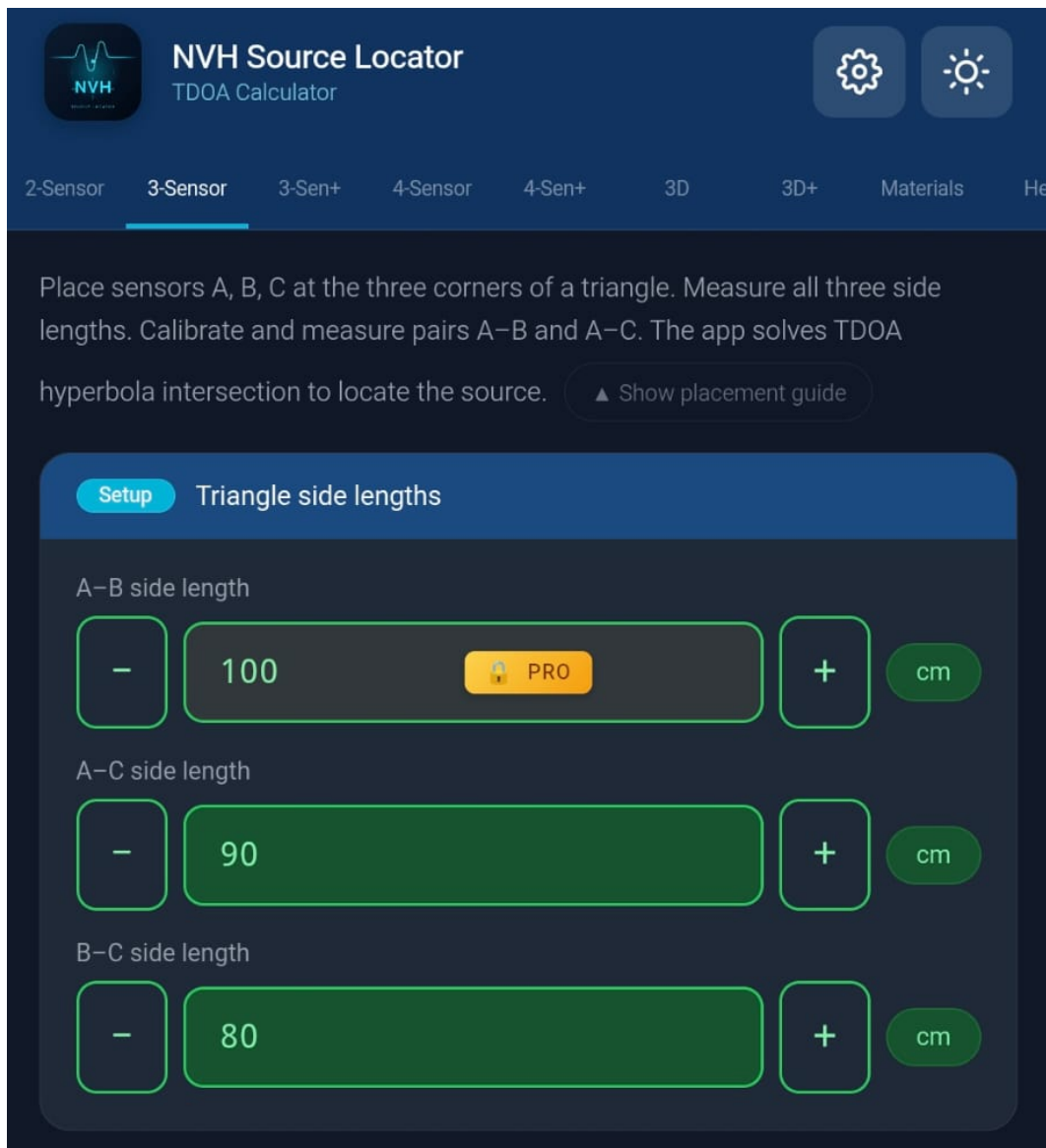
{#backup-and-restore}

, , ,

→ → " "
JSON
(Google Drive, iCloud, OneDrive),
,

→ →
, , ,

{#settings}



Paywall



- ✓ 3-Sensor & 4-Sensor triangulation
- ✓ 3D source location (X, Y, Z)
- ✓ Unlimited measurement history
- ✓ Save custom materials
- ✓ Backup and restore
- ✓ PDF reports with photo annotation
- ✓ Temperature compensation across all tabs

Looking for a discount code?
Check the Picoscope Forum →

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